



All the cell evaluations are positive and hence this affords the optimal solution which is $x_{12} = 4$, $x_{14} = 4$, $x_{24} = 4$, $x_{25} = 8$, $x_{31} = 4$ and $x_{33} = 6$; ϵ is the allocation in (1, 1) while 4 is the allocation in (3, 6) which corresponds to the dummy mill and it remains undespached at F_3 . Thus when $\epsilon \rightarrow 0$, the transportation cost is

$$4 \times 2 + 4 \times 2 + 4 \times 2 + 8 \times 1 + 4 \times 6 + 6 \times 4 + 4 \times 0 = 80.$$

EXAMPLES VIII

01 Write down the following transportation problem into a linear programming problem form :

	D_1	D_2	a_i
O_1	3	2	10
O_2	1	4	8
b_j	12	6	

[Burdwan Hons., 2010]

[Ans.] Minimize $z = 3x_1 + 2x_2 + x_3 + 4x_4$ subject to $x_1 + x_2 = 10$, $x_3 + x_4 = 8$, $x_1 + x_3 = 12$, $x_2 + x_4 = 6$, $x_1, x_2, x_3, x_4 \geq 0$.

02 (a) Find the basic feasible solution of the following transportation problems by North-West Corner rule :

(i)

	D_1	D_2	D_3	D_4	a_i
O_1	19	20	50	10	7
O_2	70	30	40	60	9
O_3	40	8	70	20	18
b_j	5	8	7	14	

(ii)

	TO			a_i
FROM	2	7	4	5
	3	3	1	8
	5	4	7	7
	1	6	2	14
b_j	7	9	18	

(iii)

	DESTINATIONS					a_i
ORIGINS	2	11	10	3	7	4
	1	4	7	2	1	8
	3	9	4	8	12	9
b_j	3	3	4	5	6	

[Jadavpur M.Sc., 1992]

(b) Determine the basic feasible solution of Ex. 2(a) (iii) by Vogel's approximation method.

- [Ans.]** (a) (i) $x_{11} = 5$, $x_{12} = 2$, $x_{22} = 6$, $x_{23} = 3$, $x_{33} = 4$, $x_{34} = 14$.
 (ii) $x_{11} = 5$, $x_{21} = 2$, $x_{22} = 6$, $x_{32} = 3$, $x_{34} = 4$, $x_{42} = 14$.
 (iii) $x_{11} = 3$, $x_{12} = 1$, $x_{22} = 2$, $x_{23} = 4$, $x_{24} = 2$, $x_{34} = 3$, $x_{35} = 6$.
 (b) $x_{14} = 4$, $x_{22} = 2$, $x_{25} = 6$, $x_{31} = 3$, $x_{32} = 1$, $x_{33} = 4$, $x_{34} = 1$.

03 Apply (a) Vogel's approximation method and

(b) Row minima method to find the basic feasible solution of the following transportation problem :

	1	2	3	4	a_i
1	21	16	25	13	11
2	17	18	14	23	13
3	32	27	18	41	19
b_j	6	10	12	15	

[Ans.] (a) $x_{14} = 11, x_{21} = 6, x_{22} = 3, x_{24} = 4, x_{32} = 7, x_{33} = 12$.
 (b) $x_{14} = 11, x_{21} = 1, x_{22} = 12, x_{31} = 5, x_{32} = 10, x_{34} = 4$.

Q4 In the previous problems, apply the matrix minima method to find the basic feasible solution in each.

Q5 Find the optimal (minimum) solution of the following transportation problems :

(i)

	D_1	D_2	D_3	D_4	a_i
O_1	1	2	1	4	30
O_2	3	3	2	1	50
O_3	4	2	5	9	20
b_j	20	40	30	10	

(ii)

	D_1	D_2	D_3	D_4	a_i
O_1	5	3	6	2	19
O_2	4	7	9	1	37
O_3	3	4	7	5	34
b_j	16	18	31	25	

[Tripura M. Sc., 1981]

(iii)

	D_1	D_2	D_3	D_4	D_5	a_i
O_1	3	4	6	8	8	20
O_2	2	10	0	5	8	30
O_3	7	11	20	40	3	15
O_4	1	0	9	14	16	13
b_j	40	6	8	18	6	

(iv)

	D_1	D_2	D_3	D_4	a_i
O_1	5	3	6	4	30
O_2	3	4	7	8	15
O_3	9	6	5	8	15
b_j	10	25	18	7	

[Kalyani M. Sc., 1991]

[Jadavpur M. Sc., 1983]

(v)

	D_1	D_2	D_3	D_4	a_i
O_1	23	27	16	18	30
O_2	12	17	20	51	40
O_3	22	28	12	32	53
b_j	22	35	25	41	

(vi)

	D_1	D_2	D_3	D_4	D_5	D_6	a_i
O_1	1	2	1	4	5	2	30
O_2	3	3	2	1	4	3	50
O_3	4	2	5	9	6	2	75
O_4	3	1	7	3	4	6	20
b_j	20	40	30	10	50	25	

[Ans.] (i) $x_{11} = 20, x_{13} = 10, x_{22} = 20, x_{23} = 20, x_{24} = 10, x_{32} = 20$;
 Min. Cost = 180.

(ii) $x_{12} = 18, x_{13} = 1, x_{21} = 12, x_{24} = 25, x_{31} = 4, x_{33} = 30$;
 Min. cost = 355.

(iii) $x_{11} = 20, x_{21} = 4, x_{23} = 8, x_{24} = 18, x_{31} = 9, x_{35} = 6, x_{41} = 7, x_{42} = 6$;
 Min. cost = 246.

(iv) $x_{12} = 23, x_{14} = 7, x_{21} = 10, x_{22} = 2, x_{23} = 3, x_{33} = 15$;
 Min. cost = 231.

(v) $x_{14} = 30, x_{21} = 5, x_{22} = 35, x_{31} = 17, x_{33} = 25, x_{34} = 11$;
 Min. cost = 2221.

(vi) $x_{11} = 20, x_{13} = 10, x_{23} = 20, x_{24} = 10, x_{25} = 20, x_{32} = 40$,
 $x_{35} = 10, x_{36} = 25, x_{45} = 20$; Min. cost = 430.]

06 Determine the optimal basic solution to the following transportation problems :

(i)

	1	2	3	4	a_i
1	2	3	11	7	6
2	1	0	6	1	1
3	5	8	15	9	10
b_j	7	5	3	2	

[Kalyani Hons., 1987]

(ii)

	1	2	3	4	5	6	a_i
1	9	12	9	6	9	10	5
2	7	3	7	7	5	5	6
3	6	5	9	11	3	11	2
4	6	8	11	2	2	10	9
b_j	4	4	6	2	4	2	

[Kalyani M. Sc., 1983]

[Ans. (i) $x_{12} = 5$, $x_{13} = 1$, $x_{23} = 1$, $x_{31} = 7$, $x_{33} = 1$, $x_{34} = 2$;
Min. cost = 100.

(ii) $x_{13} = 5$, $x_{22} = 4$, $x_{26} = 2$, $x_{31} = 1$, $x_{33} = 1$, $x_{41} = 3$, $x_{44} = 2$, $x_{45} = 4$;
Min. cost = 112.]

07 Solve the following transportation problems :

(i)

	A	B	C	a_i
I	50	30	220	1
II	90	45	170	3
III	270	200	50	4
b_j	4	2	2	

(ii)

	D_1	D_2	D_3	a_i
O_1	0	2	1	5
O_2	2	1	5	10
O_3	2	4	3	5
b_j	5	5	10	

[Calcutta Hons., 1993]

[Ans. (i) $x_{11} = 1$, $x_{21} = 3$, $x_{32} = 2$, $x_{33} = 2$; cost = 820.

(ii) $x_{13} = x_{21} = x_{22} = x_{33} = 5$; Min. cost = 35.]

08 Solve the following transportation problems :

(i)

	1	2	3	4	5	a_i
1	73	40	9	79	20	8
2	62	93	96	8	13	7
3	96	65	80	50	55	9
4	57	58	29	12	87	3
5	56	23	87	18	12	5
b_j	6	8	10	4	4	

(ii)

	a	b	c	d	e	f	a_i
A	5	3	7	3	8	5	3
B	5	6	12	5	7	11	4
C	2	1	3	4	8	2	2
D	9	6	10	5	10	9	8
b_j	3	3	6	2	1	2	17

[Ans. (i) $x_{13} = 8, x_{24} = 4, x_{25} = 3, x_{31} = 5, x_{32} = 4, x_{41} = 1, x_{43} = 2, x_{52} = 4, x_{55} = 1$; cost = 1102.]

(ii) $x_{12} = 1, x_{16} = 2, x_{21} = 3, x_{25} = 1, x_{33} = 2, x_{42} = 2, x_{43} = 4, x_{44} = 2$; cost = 103.]

10 Solve the following transportation problem to maximize the profit :

	1	2	3	4	a_i
1	15	51	42	33	23
2	80	42	26	81	44
3	90	40	66	60	33
b_j	23	31	16	30	

[Ans. $x_{12} = 23, x_{21} = 6, x_{22} = 8, x_{24} = 30, x_{31} = 17, x_{33} = 16$; Max. profit = 7005 units.]

10 Solve the following minimization transportation problem :

	1	2	3	4	5	a_i
1	4	1	3	4	4	60
2	2	3	2	2	3	35
3	3	5	2	4	4	40
b_j	22	45	20	18	30	135

[North Bengal Hons., 2008]

[Ans. $x_{12} = 45, x_{15} = 15, x_{21} = 17, x_{24} = 18, x_{31} = 5, x_{33} = 20, x_{35} = 15$;
Min. cost = 290

or $x_{12} = 45, x_{15} = 15, x_{21} = 2, x_{24} = 18, x_{25} = 15, x_{31} = 20, x_{33} = 20$.]

11 Is

	1	2	3	4
1			50	20
2	55			
3	30	35		25

an optimal solution of the following transportation problem?

	1	2	3	4	a_i
1	6	1	9	3	70
2	11	5	2	8	55
3	10	12	4	7	90
b_j	85	35	50	45	

If not, modify it to obtain a better feasible solution.

Find the solution of the modified problem.

[Ans. $x_{12} = 30, x_{14} = 40, x_{22} = 5, x_{23} = 50, x_{31} = 85, x_{34} = 5$; cost = 1160.]

[I. V. Sc., 1971]

12 A company has factories at A, B, C which supply warehouses at D, E, F, G. Monthly factory capacities are 160, 150 and 190 units respectively.

	TO			
	D	E	F	G
FROM A	42	48	38	37
B	40	49	52	51
C	39	38	40	43

Unit transportation costs (in rupees) are given in the above table.

Monthly warehouse requirements are 80, 90, 110 and 160 units respectively.

Use Modi's method to determine the optimum distribution for the company to minimize the transportation cost.

[Ans $x_{14} = 160$, $x_{21} = 80$, $x_{22} = 10$, $x_{32} = 80$, $x_{33} = 110$; Min. cost = 17,050 (rupees).]

13 A company has three plants at locations A, B and C, which supply to warehouses located at D, E, F, G and H. Monthly plant capacities are 800, 500 and 900 units respectively. Monthly warehouse requirements are 400, 400, 500, 400 and 800 units respectively. Unit transportation costs (in rupees) are given in the table below.

	TO				
	D	E	F	G	H
FROM A	5	8	6	6	3
B	4	7	7	6	6
C	8	4	6	6	3

Determine an optimal distribution for the company in order to minimize the total transportation cost.

[Delhi, 1973]

[Ans $x_{15} = 800$, $x_{21} = 400$, $x_{24} = 100$, $x_{32} = 400$, $x_{33} = 100$, $x_{34} = 300$, $x_{43} = 300$;

Min. cost = 9,200 (rupees).]

14 Find the minimum cost of transportation for the following unbalanced transportation problem (symbols have their usual meanings) :

$O_i \backslash D_j$	D_1	D_2	D_3	a_i
O_1	4	3	2	10
O_2	1	5	0	13
O_3	3	8	6	12
b_j	8	5	4	

[Calcutta Hons., 1982]

[Ans. Min. cost = 23.]

15 Consider four bases of operations B_i and three targets T_j . The tons of bombs per air craft from any base that can be delivered are given in the following tableau :

		TARGETS T_j		
		1	2	3
BASES B_i	1	8	6	5
	2	6	6	6
	3	10	8	4
	4	8	6	4

The daily sortie capability of each of the four bases is 150 sorties per day. The daily requirement in sorties over each individual target is 200. Find the allocation of sorties from each base to each target which maximizes the total tonnage over all three targets. [Roorkee, 1977]

[Ans. $x_{11} = 50, x_{12} = 100, x_{21} = 150, x_{33} = 150, x_{42} = 100, x_{43} = 50;$
 $Z_{\text{opt}} = 3,300.$] (This solution is not unique).

16 There are three sources which store a given product. These sources supply these products to four dealers. The capacities of the sources and the demands of the dealers are given as :

Capacities $S_1 = 150, S_2 = 40, S_3 = 80,$

Demands $D_1 = 90, D_2 = 70, D_3 = 50, D_4 = 60.$

	D_1	D_2	D_3	D_4
S_1	27	23	31	69
S_2	10	45	40	32
S_3	30	54	35	57

The costs of transporting products from various sources to various dealers are shown in the above tableau.

Find the optimal solution for transporting the products at a minimum cost.

[Calcutta Hons., 1984]

[Ans. $x_{11} = 30, x_{12} = 70, x_{13} = 50, x_{24} = 40, x_{31} = 60, x_{34} = 20;$
 Min. cost = 8,190.]

17 Solve the following transportation problem :

CONSUMER SUPPLIER	A	B	C	AVAILABLE
I	6	8	4	14
II	4	9	8	12
III	1	2	6	5
REQUIRED	6	10	15	31

[Baroda, 1978]

[Ans. $x_{13} = 14, x_{21} = 6, x_{22} = 5, x_{23} = 1, x_{32} = 5;$ cost = 143.]

18 Given the following data :

		DESTINATIONS			CAPACITIES
		I	II	III	
SOURCES	1	2	2	3	10
	2	4	1	2	15
	3	1	3	X	40
DEMANDS		20	15	30	

The cost of transportation from third source to the third destination is not known. How many units should be transported from the sources to the destinations so that the total cost of all the units to their destinations is a minimum.

[Assume a huge cost M in the cell (3, 3). Then apply VAM.]

[Ans. $x_{12} = 10, x_{23} = 15, x_{31} = 20, x_{32} = 15$ and $x_{33} = 5.$]



19 A company has four warehouses a, b, c, d . It is required to deliver a product from these warehouses to three customers A, B, C . The warehouses have the following amounts in stock :

Warehouse	a	b	c	d
No. of units	15	16	12	13

and the customers' requirements are

Customer	A	B	C
No. of units	18	20	18

The table below shows the cost of transporting one unit from warehouses to the customers

	a	b	c	d
A	8	9	6	3
B	6	11	5	10
C	3	8	7	9

Solve the transportation problem.

[Bombay, 1977]

[Ans. $x_{12} = 5, x_{14} = 13, x_{22} = 8, x_{23} = 12, x_{31} = 15, x_{32} = 3$; Min. cost = 301.]

20 Hindusthan Construction Company needs 3, 3, 4 and 5 million cubic feet of fill at four earthen dam sites in Punjab. It can transfer the fill from three mounds A, B, C where 2, 6 and 7 million cubic feet of fill is available, respectively. Costs of transporting one million cubic feet of fill from the mounds to the four sites in lakhs are given by

		TO DAM SITES			
		I	II	III	IV
FROM MOUNDS	A	15	10	17	18
	B	16	13	12	13
	C	12	17	20	11

Solve the problem using transportation algorithm for minimum cost.

[Ans. $x_{12} = 2, x_{22} = 1, x_{23} = 4, x_{24} = 1, x_{31} = 3, x_{34} = 4$; Min. cost = 174.]

21 Consider the following unbalanced transportation problem :

		TO			
		1	2	3	
FROM	I	5	1	7	10
	II	6	4	6	80
	III	3	2	5	15
		75	20	50	

Since there is not enough supply, some of the demands at these destinations may not be satisfied. Suppose there are penalty costs for every unsatisfied demand unit which are given by 5, 3, 2 for destinations 1, 2, 3 respectively. Find the optimal solution.

[For the dummy source, assume transportation costs 5, 3, 2 and $145 - 105 = 40$ as its supply. Then proceed.]

[Ans. $x_{12} = 10, x_{21} = 60, x_{22} = 10, x_{23} = 10$ and $x_{31} = 15$;
cost = 515 + penalty 80 = 595.]

22 There are three parties who supply the following quantities of coal and three consumers who require the coal as follows :

Party 1	14 tons
Party 2	12 tons
Party 3	5 tons
Total	31 tons

Consumer A	6 tons
Consumer B	10 tons
Consumer C	15 tons

	A	B	C
1	6	8	4
2	4	9	3
3	1	2	6

The cost matrix is as shown above. Find the schedule of a transportation policy which minimizes the cost.

[Ans. $x_{12} = 5$, $x_{13} = 9$, $x_{21} = 6$, $x_{23} = 6$, $x_{32} = 5$; cost = 128.]

23 Four products are produced in three machines and their profit margins (machine to product) on sale are given in the following table :

		PRODUCTS				
		P_1	P_2	P_3	P_4	CAPACITIES
MACHINES	M_1	6	4	1	5	14
	M_2	8	9	2	7	18
	M_3	4	3	6	2	7
DEMANDS		6	10	15	8	

Demands of the products and capacities of the machines per week are shown in the table. Find a suitable production plan of the products in the machines to maximize the profit. Find any other optimal solution to the problem, if there be any.

[Ans. $x_{13} = 8$, $x_{14} = 6$, $x_{21} = 6$, $x_{22} = 10$, $x_{24} = 2$, $x_{33} = 7$; Max. profit = 232 units
or $x_{11} = 6$, $x_{13} = 8$, $x_{22} = 10$, $x_{24} = 8$, $x_{33} = 7$.]

24 A manufacturer has distribution centres located at Agra, Allahabad and Calcutta. Those centres have available 40, 20 and 40 units of his product. His retail outlets require the following number of units :

A = 25, B = 10, C = 20, D = 30 and E = 25.

The cost of transportation per unit in rupees between each centre and outlet is given in the table below :

DISTRIBUTION CENTRES	RETAIL OUTLETS				
	A	B	C	D	E
AGRA	55	30	40	50	40
ALLAHABAD	35	30	100	45	60
CALCUTTA	40	60	95	35	30

(i) Determine the optimal transportation cost.

(ii) Is there any alternative solution?

(iii) Find the idle capacity of the retail outlets, if any.

[Ans. (i) $x_{12} = 10$, $x_{13} = 20$, $x_{15} = 10$, $x_{21} = 20$, $x_{34} = 25$, $x_{35} = 15$; Opt. cost = 3525.

(ii) No alternative optimal solution.

(iii) 5 units at A and 5 units at D.]

25 Find an optimal solution and corresponding cost of transportation in the following transportation problem :

	D_1	D_2	D_3	D_4	
O_1	1	2	-2	3	70
O_2	2	4	0	1	38
O_3	1	2	-2	5	32
	40	28	30	42	

[Calcutta Hons., 2005]

[Ans. $x_{11} = 40$, $x_{12} = 26$, $x_{14} = 4$, $x_{24} = 38$, $x_{32} = 2$, $x_{33} = 30$; Min. cost = 86.]

[The negative cost may be avoided by adding 2 to all cell values of the cost matrix.]